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...thanks to...

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Abstract

Precision agriculture relies heavily on the accurate, large-scale mapping of land use and crop distribution derived from Earth Observation satellites, particularly the high-resolution, high-frequency multispectral imagery provided by the Sentinel-2 mission. By monitoring how specific agricultural parcels reflect light over time, researchers construct Satellite Image Time Series (SITS) to track the unique phenological growth stages of varying vegetation. Despite these advancements, employing deep learning for SITS on a national or global scale presents severe technical and agronomic challenges. Satellite imagery is inherently irregular due to persistent cloud cover and atmospheric interference, and processing high-dimensional spatial-temporal grids creates critical memory and input/output (I/O) bottlenecks in High-Performance Computing (HPC) environments. Furthermore, many current state-of-the-art sequence models suffer from "temporal myopia". They treat each growing season as an entirely isolated event, attempting to classify crops based on just one year of weather and growth patterns. In doing so, they completely ignore a fundamental rule of farming: crop rotation. Farmers intentionally alternate specific crops across consecutive years to maintain soil health and prevent nutrient depletion. If a satellite observation is heavily obscured by clouds or spectrally ambiguous in a given year, a model suffering from temporal myopia has no historical context to help it classify the vegetation. Conversely, a model capable of tracking multi-year rotations can analyze what the farmer planted in the previous and subsequent years to logically deduce the missing crop.

This thesis proposes a highly scalable machine learning framework designed to solve these hardware limitations while simultaneously capturing both intra-year phenology and inter-year crop rotation dynamics. To overcome the severe HPC bottlenecks associated with large-scale spatial processing, the framework transitions the data pipeline to a highly optimized **1D pixel-based extraction method**. By reducing the data representation to the deep temporal evolution of individual pixels, the architecture bypasses memory limits and enables the rapid processing of a massive six-year continuous SITS dataset over Austria (INVEKOS). To support rigorous architectural evaluation, the project utilizes an automated Machine Learning Operations (MLOps) pipeline integrating PyTorch Lightning, Ray Tune, and Optuna to guarantee fair hyperparameter tuning. Additionally, to handle irregular temporal gaps specifically within the Transformer models, the framework introduces a custom **AgroDays Positional Encoding** module, which scales sinusoidal encodings using the exact agricultural day of the satellite acquisition to explicitly inject temporal distance into the spectral embeddings.

The research systematically establishes robust single-year baselines to serve as standard benchmarks, including a **1-Dimensional Convolutional Neural Network (1D-CNN)** and standalone **Transformer Encoders** evaluated in both generative and discriminative variants. Crucially, the research also introduces **linear structural controls**; in these tests, the deep non-linear feature extractors are replaced by a simplistic linear backbone that feeds directly into multi-year sequence heads. This deliberate structural control definitively proves the mathematical necessity of utilizing high-capacity, non-linear neural backbones before multi-year sequence smoothing can be applied to raw satellite data.

To bridge the gap between phenological tracking and historical agricultural dependencies, two distinct multi-year hybrid architectures are confronted. The established **Transformer-HMM cascade** acts as the Bayesian reference, where the Transformer estimates yearly class probabilities and a Hidden Markov Model (HMM) smooths these predictions over time using explicit, fixed transition matrices. Against this, the thesis proposes a novel **Hierarchical Transformer-LSTM**, a purely neural end-to-end architecture. It processes intra-year sequences in parallel using self-attention, and feeds the chronological yearly predictions into an inter-year bidirectional LSTM. This forces the network to learn multi-year crop rotation rules from past and future planting contexts without the use of predefined

agronomic matrices.

Experimental results across 47 distinct crop classes demonstrate the severe limitations of single-year models on highly imbalanced datasets. The 1D-CNN and standalone Discriminative Transformer achieved limited Macro F1-scores of 30.17% and 33.18%, respectively, while the linear structural controls suffered catastrophic performance degradation. Ultimately, both multi-year architectures vastly outperformed the single-year baselines, demonstrating that historical crop rotation context is highly effective at rescuing minority classes that traditional models ignore. The novel Transformer-LSTM successfully learned complex agricultural cycles end-to-end, achieving classification performance competitive with the state-of-the-art HMM cascade. By delivering this optimized framework, this thesis provides a highly scalable software solution for operational crop mapping, laying the groundwork for future research which might integrate State Space Models, Graph Neural Networks, and multimodal SAR data fusion.

1 Introduction and Background

1.1 Introduction

The global agricultural sector is currently facing unprecedented challenges. Driven by rapid population growth, climate change, and the need for sustainable resource management, modern agriculture must maximize crop yields while minimizing environmental impact [1]. Precision agriculture [2, 3] has emerged as a crucial solution to address these demands, relying heavily on accurate, large-scale, and up-to-date information regarding land use and crop distribution. Traditionally, acquiring this data required costly and time-consuming manual field surveys, which are impossible to scale effectively at a national or global level.

Over the past decade, Earth Observation (EO) satellites have revolutionized how we monitor agricultural lands. Constellations such as the European Space Agency's Sentinel-2 provide free, high-resolution multispectral imagery with frequent revisit times (every 5 to 10 days) [4]. By observing how a specific parcel of land reflects light over time [5], it is possible to track the phenological cycle—the biological growth stages—of vegetation. This continuous stream of data allows researchers to identify specific crop types based on their unique growth signatures rather than relying on a single, static snapshot [6].

Simultaneously, the rise of Deep Learning (DL) has provided the computational tools necessary to process this massive influx of satellite data. Early approaches in remote sensing relied heavily on traditional machine learning algorithms and spatial pattern recognition. However, as the field evolved, it became evident that the temporal dimension—how a pixel changes over a year—holds the most predictive power for crop classification. Consequently, sequence-modeling architectures such as Temporal Convolutional Neural Networks (1D-CNNs) [7, 8] and Recurrent Neural Networks (RNNs) [9] have become the standard for agricultural time-series analysis.

Despite these advancements, operationalizing deep learning for crop mapping presents significant difficulties. First, Satellite Image Time Series (SITS) are inherently irregular; optical sensors cannot penetrate clouds, leading to missing data and uneven temporal gaps that disrupt traditional machine learning models. Second, processing high-dimensional spatial-temporal datasets—such as tracking entire 2D image patches over a temporal sequence—creates severe computational and memory bottlenecks, particularly in High-Performance Computing (HPC) environments. Finally, most current state-of-the-art models suffer from **"temporal myopia"**; they analyze only a single year of weather and phenology, completely ignoring the fundamental agricultural practice of **crop rotation**, where farmers intentionally alternate crops across multiple years to maintain soil health [10].

This thesis aims to bridge these gaps. To overcome the hardware bottlenecks associated with heavy spatial-temporal grids, this work shifts to a highly optimized **1D pixel-based pipeline**. By analyzing the temporal evolution of individual pixels rather than broad spatial patches, the computational load is drastically reduced, enabling rapid prototyping and extensive hyperparameter optimization on an HPC cluster. Building upon this efficient infrastructure, the project establishes a comprehensive suite of robust baselines. Single-year intra-year models—specifically 1D-CNNs and standalone Transformer Encoders—are tuned to serve as standard phenological benchmarks. Alongside these, simplified baseline models pairing a basic linear layer directly with standalone LSTM [11] and HMM [12] heads are evaluated. These configurations serve as a deliberate benchmark test, demonstrating that high-capacity, non-linear feature extractors are strictly necessary for sequence models—in particular, the multi-year probabilistic smoothing performed by the HMM—to be able to effectively interpret raw satellite data.

These baselines are systematically compared against two high-capacity, multi-year hybrid architectures. The first is the **Transformer-HMM cascade**, an established state-of-the-art Bayesian

framework [13] utilized here as an advanced reference model. To optimize the emission probabilities provided to the HMM transition layer, different backbone architectures are explicitly compared—including 1D-CNNs and both generative and discriminative variants of the Transformer backbone [13]. Although state-space models like Mamba [14] were originally slated for this feature-extraction comparison, time constraints limited the executed backbones to convolutions and self-attention. The second is a novel, purely neural end-to-end architecture proposed by this thesis: the **Hierarchical Transformer-LSTM**. By integrating the self-attention mechanism proven effective on satellite time series [15, 16] with the recurrent memory of an LSTM, this research systematically confronts Bayesian transition modeling against deep recurrent memory to answer whether neural networks can intrinsically learn and leverage the historical rules of multi-year crop rotation to significantly outperform models that rely on single-year phenology alone.

1.2 Research Question and Objectives

The goal of this thesis is to explore the efficacy of deep sequence-modeling architectures in classifying agricultural crop types using irregular SITS. Specifically, this research investigates the transition from traditional, single-year phenological analysis to a multi-year, hybrid approach capable of learning long-term farming practices. This leads to the primary research question of this study:

> *"How can we efficiently process irregular Satellite Image Time Series for large-scale crop mapping, and does the integration of multi-year crop rotation history via hybrid Transformer-LSTM and Transformer-HMM architectures significantly outperform traditional single-year phenological models?"*

To systematically answer this question, the project is structured around four specific objectives, spanning both machine learning infrastructure and architectural design:

i) **Overcoming I/O and Computational Bottlenecks:** To redesign the data pipeline, shifting from heavy spatial-temporal patches to a highly optimized 1D pixel-based extraction method. This aims to resolve severe High-Performance Computing (HPC) memory constraints and reduce epoch training times, enabling rapid experimentation.

ii) **Establishing Robust Control Baselines:** To design, train, and tune standard non-hybrid temporal architectures to serve as phenological benchmarks. Specifically, a 1-Dimensional Convolutional Neural Network (1D-CNN) and standalone Transformer Encoders—evaluated in both generative and discriminative variants—are explicitly compared to determine optimal backbone feature extraction capabilities. While state-space models such as Mamba were initially intended to be part of this backbone comparison, time and computational constraints precluded their execution. Alongside these, standalone LSTM and HMM configurations utilizing a simplistic linear backbone are executed. These provide a deliberate structural control test, demonstrating the absolute necessity of high-capacity non-linear feature extractors for raw temporal data before multi-year sequence smoothing can be applied.

iii) **Developing Multi-Year Hybrid Architectures:** To construct the novel Hierarchical Transformer-LSTM and confront it against the established Transformer-HMM cascade to evaluate sequence modeling beyond a single harvest cycle. This objective focuses on using self-attention to manage intra-year weather irregularities, while comparing an LSTM memory cell against an HMM cascade to capture inter-year crop rotation dependencies. Additionally, both generative and discriminative variants of the Transformer backbone are tested to optimize the emission probabilities passed to the sequence heads.

iv) **Automating Hyperparameter Optimization:** To build a robust, fully automated Machine Learning Operations (MLOps) pipeline integrating PyTorch Lightning, Ray Tune, and Optuna. This ensures all models are evaluated fairly at their peak capacity through distributed hyperparameter tuning on an HPC cluster.

1.3 State of the Art

1.3.1 Deep Learning for Precision Agriculture

The paradigm of precision agriculture has fundamentally transformed modern farming management. By transitioning from uniform field-level treatments to highly granular, site-specific interventions, precision agriculture seeks to optimize resource inputs—such as water, fertilizers, and pesticides—while maximizing overall crop yield [2, 3]. This shift relies entirely on the continuous acquisition and analysis of vast amounts of heterogeneous data collected from on-site Internet of Things (IoT) sensors, Unmanned Aerial Vehicles (UAVs), and Earth Observation (EO) satellites.

Historically, extracting actionable insights from these diverse data sources depended on traditional machine learning algorithms, such as Random Forests (RF) and Support Vector Machines (SVM) [17, 18]. While effective for specific, localized tasks, these traditional methods are inherently limited by their reliance on manual feature engineering. Researchers were required to construct and select specific vegetation indices or spatial textures beforehand, a process that is highly domain-specific, time-consuming, and difficult to scale across different geographical regions and climatic conditions.

The advent of Deep Learning (DL) has largely overcome these limitations, triggering a major shift in agricultural data analysis [19]. Unlike traditional machine learning, deep neural networks possess the capacity for automatic, hierarchical feature extraction. By ingesting raw or minimally processed data, DL models can autonomously learn complex, non-linear representations directly from the input space. In precision agriculture, this has led to unprecedented breakthroughs across a wide array of applications, including automated weed detection, early disease diagnosis, and precise yield estimation [20].

In the specific context of remote sensing and land cover mapping, deep learning has proven exceptionally proficient at handling the massive dimensionality of Satellite Image Time Series (SITS). By leveraging specialized architectures capable of modeling both spatial hierarchies and temporal dependencies, deep learning models can now accurately map crop distributions at a national scale, outperforming traditional shallow classifiers and forming the foundational technology upon which this thesis is built [21].

1.3.2 Remote Sensing and Sentinel-2 Time Series

The Importance of Spatio-Temporal Dynamics

The European Space Agency's Sentinel-2 mission has revolutionized agricultural monitoring by providing high spatial resolution imagery with a frequent 5-day revisit time. This creates dense Satellite Image Time Series (SITS) that capture the vital phenological evolution of crops throughout the growing season [22]. However, extracting actionable insights from SITS requires models to navigate a complex time-space tradeoff. Research demonstrates that for crop type classification on Sentinel-2 data, the temporal structure is often richer than the spatial structure; consequently, hybrid models (like recurrent convolutional networks) perform best when the majority of their parameters are allocated to modeling temporal sequences rather than spatial features [22].

Encoding and Sequence Modeling

To effectively process these time series, researchers have explored various representation and sequence-modeling strategies:

- **Image-Based Encodings:** One approach involves transforming 1D time-series data (such as vegetation indices) into 2D image representations—like Gramian Angular Fields or Markov Transition Fields—allowing standard convolutional networks to efficiently extract static and dynamic temporal patterns [23].

- **Recurrent to Transformer Evolution:** While models like Long Short-Term Memory (LSTM) networks have traditionally handled sequential data, modern approaches increasingly rely on Transformer architectures to overcome LSTM limitations with long-range dependencies [24]. In crop mapping, advanced models like the Temporo-Spatial Vision Transformer (TSVIT) utilize purely self-attention mechanisms to process multi-modal SITS, combining optical data (Sentinel-2) with radar (Sentinel-1) to boost classification performance without relying on standard convolutions [25].

Overcoming Noise and Label Scarcity

Despite the rich information in SITS, operational mapping is frequently hindered by cloud cover noise and a severe lack of pixel-level ground truth data. Recent deep learning innovations address these bottlenecks directly:

- **Knowledge-Guided Networks:** To ensure models learn meaningful phenology rather than dataset artifacts, prior domain knowledge can be integrated into the architecture. For instance, the Knowledge-Guided Crop Mapping Network (KGCMNet) forces the model to reconstruct known vegetation indices (like NDVI) during training, which significantly sharpens its spatio-temporal representations [26].
- **Weakly Supervised Learning:** To bypass the need for exhaustive pixel-level annotations, weakly supervised frameworks (e.g., Exact) utilize only image-level labels. These models explore space-time perceptive clues to rectify erroneous semantic biases caused by anomalous temporal clips, pushing weakly supervised performance to nearly 95% of fully supervised baselines [27].
- **Multimodal Pseudo-Labeling:** Another highly scalable solution involves integrating top-down SITS with bottom-up, ground-level field images. By using multimodal data to generate reliable pseudo-labels, models can achieve high accuracy across vast geographical areas while drastically reducing the need for expensive on-site data collection [28].

1.3.3 Sequence Modeling: From CNNs to Hybrid Architectures

The accurate classification of agricultural land relies heavily on the ability to interpret the chronological sequence of crop phenology. As deep learning has evolved, so too have the architectures designed to model these complex temporal dynamics.

Convolutional Neural Networks (CNNs)

Early deep learning approaches for Satellite Image Time Series (SITS) relied heavily on Convolutional Neural Networks (CNNs). While originally designed to extract spatial features from static images [20], CNNs were successfully adapted for temporal sequences via 1-Dimensional (1D) convolutions. By operating filters across the temporal dimension of a pixel's profile, 1D-CNNs can efficiently capture localized temporal patterns and phenological shifts, significantly outperforming traditional machine learning classifiers like Random Forests [21]. However, standard CNNs lack an inherent mechanism to model strict, long-term sequential dependencies, as their receptive field is limited by the kernel size.

Recurrent Neural Networks (RNNs) and LSTMs

To explicitly model the chronological order of agricultural events, Recurrent Neural Networks (RNNs) became the standard. The foundational RNN Encoder-Decoder architecture allowed models to ingest variable-length sequences and maintain a hidden state that summarizes past inputs to inform future predictions [29]. To address the vanishing gradient problem inherent in basic RNNs, Long Short-Term Memory (LSTM) networks were widely adopted. These architectures are particularly advantageous in remote sensing, as specialized recurrent layers can be engineered to handle the irregular time gaps, cloud occlusions, and missing values that commonly afflict satellite time series [30].

The Transformer Paradigm

Despite their success, LSTMs process data sequentially, which bottlenecks training parallelization and limits their ability to capture extremely long-range dependencies across a full growing season [24]. The introduction of Transformer architectures marked a paradigm shift in sequence modeling. By dispensing with recurrence and relying entirely on self-attention mechanisms, Transformers evaluate the relationships between all time steps simultaneously, regardless of their distance in the sequence [24]. In modern crop mapping applications, Transformers have demonstrated superior classification accuracy and generalizability compared to 1D-CNNs, especially when handling complex, multi-year, or irregularly sampled temporal features [31].

Crucially, the output formulation of the Transformer plays a significant role when interfacing with probabilistic sequence models like HMMs. Transformers can be configured discriminatively (using a standard linear layer with softmax) to estimate class posterior probabilities directly, or generatively (using a normalized softmax layer with no bias term) to implicitly enforce a uniform class prior [13].

The generative approach reduces bias toward dominant classes and aligns perfectly with the emission model requirements of Bayesian transition networks [13].

Hybrid Architectures

To maximize predictive performance, researchers have increasingly developed hybrid architectures that integrate multiple sequence-modeling paradigms. Early hybrid networks in remote sensing frequently combined CNNs with RNNs (such as ConvLSTMs or CNN+GRU architectures) to simultaneously extract spatial textures and temporal phenology [22]. As the focus has shifted toward capturing highly complex, multi-scale temporal dynamics, newer state-of-the-art approaches merge the localized sequential extraction capabilities of LSTMs with the global, long-range processing power of Transformers [32, 24].

While these standard hybrid models represent the current benchmark for decoding spatiotemporal Earth observation data, they typically process sequences within a single temporal scale. This thesis proposes a novel inversion of this paradigm specifically tailored for multi-year agricultural monitoring. Rather than using both networks on the same sequence, the proposed architecture separates the timescales: Transformers are utilized to concurrently extract dense, intra-year phenological features, while Bidirectional LSTMs (as proposed by this thesis) or Hidden Markov Models (as established by recent literature) are subsequently deployed to capture the long-term, inter-year sequential dependencies of crop rotations.

1.3.4 Multi-Year Crop Rotation Modeling

The Importance of Long-Term Agricultural Dynamics

Crop rotation is a fundamental agricultural practice wherein farmers alternate the species of crops grown on a specific plot across consecutive seasons. This strategy is essential for maintaining soil health, preventing nutrient depletion, and breaking pest and disease cycles. Despite its agronomic importance, most remote sensing models suffer from "temporal myopia," analyzing single-year phenology while ignoring inter-annual historical dependencies.

Traditional and Knowledge-Based Approaches

Early attempts to map multi-year dynamics relied on explicitly injecting prior agronomic knowledge into the classification pipeline. For example, probabilistic models such as Bayesian networks and Conditional Random Fields have been used to encode known crop rotation rules; the outputs of these networks are then fed into standard classifiers (such as Support Vector Machines) to guide the algorithm and improve early-season mapping accuracy [17, 22]. While effective, these methods require manually constructed rule bases and lack the flexibility to adapt to unexpected crop sequence changes.

Machine Learning and Temporal Transferability

As satellite archives have grown, researchers have increasingly utilized machine learning to track land-use transitions over longer periods. Hierarchical classification workflows (such as tiered Random Forests) have been successfully deployed across multiple years to explicitly track crop rotations, field abandonment, and land-use intensity [18]. However, monitoring annual crop rotations over a decade presents significant challenges regarding temporal transferability; models trained on a specific year often struggle to generalize to past or future years due to shifting weather patterns, varying sensor availability, and inter-annual phenological shifts [31].

The Deep Learning Frontier

While early deep learning methods successfully tracked multi-year changes, they predominantly processed each year independently and relied on post-classification comparisons to identify rotations. The next frontier in precision agriculture mapping is the development of architectures that possess the intrinsic memory capacity to learn and predict multi-year agricultural cycles directly from continuous, raw satellite time-series data.

Recently, researchers have demonstrated significant success in bridging these paradigms by integrating Bayesian modeling—such as Hidden Markov Models (HMMs)—with deep Transformer Encoders to explicitly capture intricate multi-year crop rotation patterns [13]. In this cascade setup, the Transformer acts as the emission model providing class likelihoods, while the HMM acts as the transition model. The HMM leverages explicit transition matrices—probabilistic grids that mathematically quantify the likelihood of a farmer switching from one specific crop type to another across consecutive seasons [12, 13]. Building upon these foundations, the focus now shifts toward confronting these probabilistic

cascades with purely neural hybrid architectures, such as LSTMs, capable of modeling complex temporal dependencies end-to-end without relying on fixed, explicit transition matrices.

1.4 Contributions

This thesis advances the field of agricultural remote sensing by addressing the computational bottlenecks of large-scale mapping and exploring the predictive power of multi-year sequence modeling. The primary contributions of this work are as follows:

- **Adoption of an Optimized 1D Data Pipeline:** To overcome the limitations of memory-intensive 2D spatial patches, a streamlined, 1D pixel-based data representation is utilized. This approach drastically reduces High-Performance Computing (HPC) I/O bottlenecks, enabling the rapid processing and evaluation of massive, multi-year Satellite Image Time Series (SITS).
- **Comprehensive Baseline Benchmarking:** Robust single-year temporal baseline models—specifically a **1-Dimensional Convolutional Neural Network (1D-CNN)** and standalone **Transformer Encoders** (evaluated in both generative and discriminative variants)—are developed and tuned to provide a standard for intra-year phenological classification. Additionally, standalone tests using a simplified **linear backbone** paired with **LSTM** and **HMM** heads are executed. These serve as a deliberate control mechanism, proving the structural necessity of high-capacity non-linear feature extractors when processing raw SITS.
- **Confronting Novel and Established Multi-Year Architectures:** A custom Hierarchical **Transformer-LSTM** architecture is designed and compared to the established **Transformer-HMM** Bayesian cascade. This comparison evaluates whether deep recurrent memory or probabilistic transition matrices better capture complex inter-year crop rotation dependencies. Furthermore, both generative and discriminative variants of the Transformer are evaluated to determine the optimal configuration for generating emission probabilities.
- **Integration of Automated MLOps and Distributed Tuning:** A robust Machine Learning Operations (MLOps) workflow, leveraging PyTorch Lightning, Ray Tune, and Optuna, is adapted to automate the training pipeline. This infrastructure facilitates distributed hyperparameter optimization across an HPC cluster, ensuring that all architectures are evaluated fairly and at their maximum potential.

By delivering this end-to-end framework, this thesis not only provides a scalable software solution for operational crop mapping but also offers critical insights into whether neural networks can effectively learn the historical, multi-year rules of agricultural crop rotation.

2 Methodology and Architectural Design

2.1 Single-Year and Structural Baselines

To establish a rigorous benchmark for multi-year architectures, a comprehensive suite of single-year sequence models and structural controls was developed. These networks are restricted to processing intra-year phenological data, providing a comparative standard to justify the complexity of multi-year modeling.

2.1.1 Temporal 1D Convolutional Neural Network (1D-CNN)

The first phenological baseline is a 1-Dimensional Convolutional Neural Network (1D-CNN). By sliding convolutional filters across the time dimension of a pixel's spectral profile, the network efficiently extracts localized temporal features and phenological shifts, a methodology that has proven highly effective for satellite time series classification [33].

The **encoder** consists of two sequential convolutional blocks. The initial layer maps the 13 input satellite bands to 64 channels using a kernel size of 3 and a padding of 1 to preserve the temporal sequence length. This is followed by a 1D Batch Normalization layer and an in-place ReLU activation to introduce non-linearity. The second layer further projects the features from 64 to 128 channels, utilizing the same kernel size, padding, normalization, and activation parameters.

For **classification**, a point-wise convolution (kernel size of 1) is applied to map the extracted 128-dimensional features to the 47 target crop classes. Because the network produces predictions for each time step, a temporal majority vote is executed by averaging the unnormalized logit scores across all time steps in the year to generate the final field-level classification.

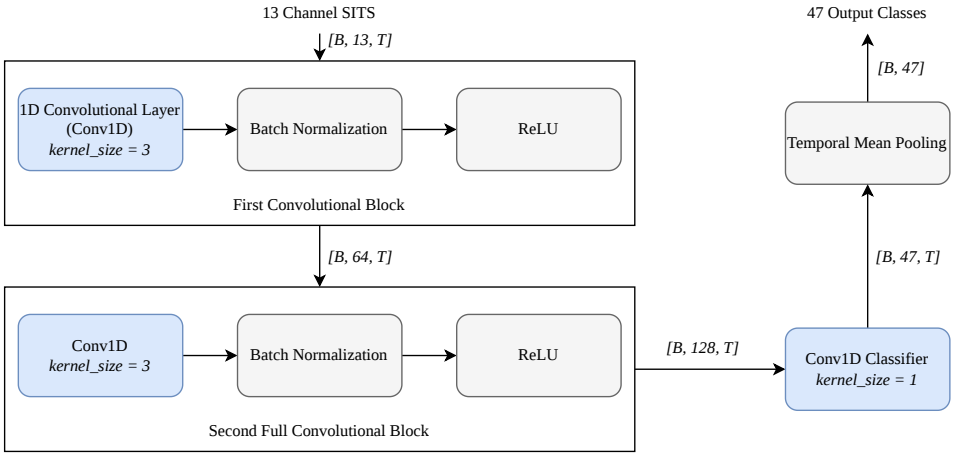


Figure 2.1: Architectural block diagram of the Temporal 1D-CNN baseline.

2.1.2 Standalone Transformer Encoders (Generative and Discriminative)

To capture global temporal dependencies without being restricted to the narrow, sliding time windows of a CNN, a standalone **Transformer Encoder** serves as the second primary baseline. Utilizing Multi-Head Self-Attention [15], this architecture evaluates the entire yearly sequence simultaneously, mapping the complex growth stages of crops into dense representations [34].

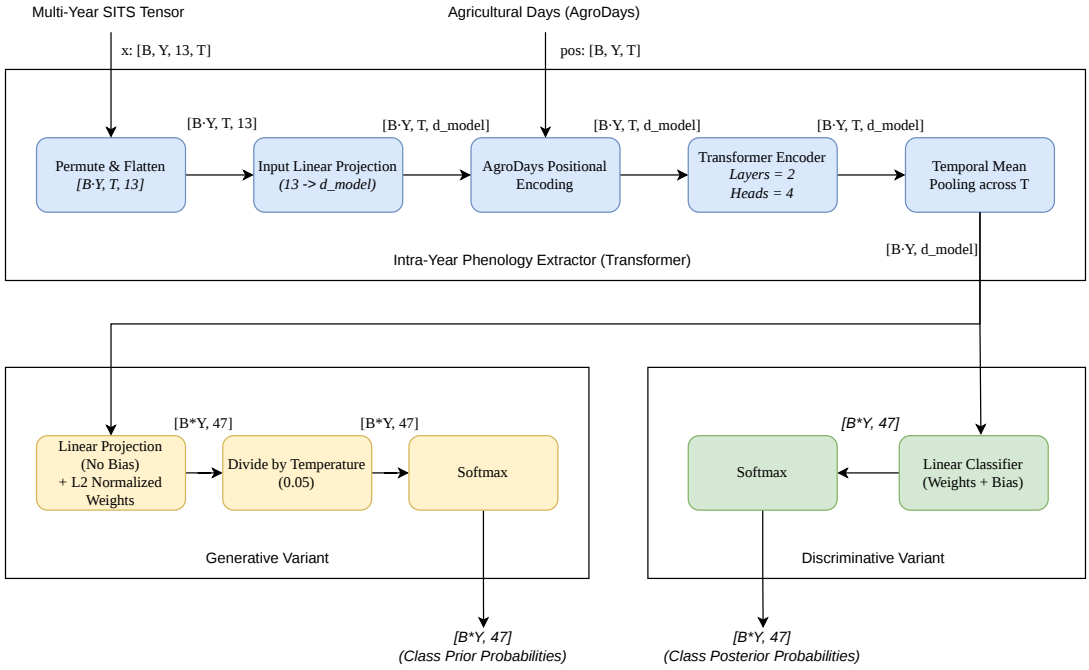


Figure 2.2: Architectural block diagram of the standalone Transformer Encoder (Generative and Discriminative variants).

Crucially, the output formulation of the Transformer is evaluated in two distinct variants to benchmark their efficacy as both standalone classifiers and future emission models:

- **Discriminative Variant:** Employs a standard classification layer to directly predict the probability of each crop class based on the input sequence.
- **Generative Variant:** Utilizes a specifically modified, normalized output layer. This design treats all crop types as equally likely by default, which prevents the network from unfairly favoring the most common crops in the dataset [13]. More importantly, it generates these probabilities in the exact mathematical format required by the HMM to effectively map sequence transitions [13].

2.1.3 Linear Structural Controls

To scientifically validate the necessity of the aforementioned non-linear backbones, standalone multi-year controls utilizing a **simplistic linear backbone** were evaluated. In these configurations, the raw 13-band spectral time series is projected through a single, uninitialized linear layer ($Wx + b$) directly into the sequence-modeling heads (LSTM and HMM).

These linear models serve as a deliberate **structural simplification**. Because a single linear layer cannot mathematically separate complex, non-linear phenological profiles, it outputs near-random temporal features. Testing the sequence heads on these impoverished features isolates the capabilities of the recurrent and probabilistic networks, demonstrating that multi-year sequence smoothing fundamentally requires a high-capacity, non-linear feature extractor to function effectively.

2.2 Multi-Year Architectures

The core focus of this thesis is the development and evaluation of hybrid, multi-scale neural architectures capable of simultaneously modeling intra-year phenological dynamics and inter-year crop rotation patterns.

2.2.1 AgroDays Positional Encoding

Agricultural satellite time series are inherently irregularly sampled due to cloud cover and varying satellite revisit times. Standard sequential positional encodings would falsely inform a self-attention

mechanism that all observations are equidistant. To address this, the architecture utilizes a custom **AgroDays Positional Encoding** module.

Instead of indexing observations as sequential integer steps, this module adapts the standard sinusoidal positional encoding [15] by scaling the frequencies using the exact agricultural day of the acquisition. Mapping absolute day values onto continuous periodic functions ensures that the mathematical distance between any two encoded time steps is strictly proportional to the actual temporal gap between them.

Algorithm 1 AgroDays Positional Encoding (Lookup Table)

Require: Maximum agricultural day $max_seq \leftarrow 365$
Require: Pre-compute during initialization:
$table \leftarrow \text{zeros}(max_seq, d_model)$
for $pos = 0$ **to** max_seq **do**
  for $i = 0$ **to** $d_model/2$ **do**
    $freq \leftarrow pos/10000^{2i/d_model}$
    $table[pos, 2i] \leftarrow \sin(freq)$
    $table[pos, 2i+1] \leftarrow \cos(freq)$
  end for
end for
$PE_Layer \leftarrow \text{FrozenEmbedding}(table)$
Require: Forward Pass:
Require: Input sequence $X \in \mathbb{R}^{B \times T \times d_model}$
Require: Agricultural days tensor $pos \in \mathbb{R}^{B \times T}$
Ensure: Position-encoded sequence $X_{enc} \in \mathbb{R}^{B \times T \times d_model}$
$PE \leftarrow PE_Layer(pos)$ ▷ Look up pre-computed vectors using day index
$X_{enc} \leftarrow X + PE$ ▷ Inject temporal signal into embeddings
return X_{enc}

As detailed in Algorithm 1, the position-aware embeddings are pre-computed during the network's initialization. A geometric progression of wavelengths is calculated across the network's embedding dimension and stored within a frozen lookup table [15]. During the forward pass, the model simply uses the input matrix *pos* as indices to extract the correct sine and cosine vectors for each specific agricultural day, injecting this explicit temporal signal directly into the spectral embeddings.

2.2.2 The Reference Transformer-HMM Cascade

To evaluate purely neural multi-year models, an established state-of-the-art framework is required for comparison. This research implements the Bayesian cascade architecture proposed by Perantoni et al. [13], which integrates a Transformer Encoder with a **Hidden Markov Model** (HMM).

In this hierarchical setup, the Transformer acts exclusively as the **emission model**, processing the intra-year SITS to estimate the likelihood of each crop class for a given year. The generated class probabilities are then passed into the HMM transition layer. The HMM models the temporal correlation of the labels across the multi-year sequence utilizing explicit **transition matrices**, probabilistic grids that mathematically quantify the likelihood of a farmer rotating from one specific crop type to another. By performing cascade classification in both the forward and backward temporal directions, the HMM smooths the Transformer's emission predictions, ensuring the final predicted sequence adheres to logical agronomic rotation practices [13].

2.2.3 The Novel Hierarchical Transformer-LSTM

As an alternative to Bayesian transition smoothing, this thesis proposes a novel, purely neural multi-year architecture: the Hierarchical Transformer-LSTM. This model is designed to learn both intra-year phenology and inter-year crop rotations end-to-end, without relying on fixed, explicit transition matrices.

The architecture processes the temporal scales using a two-stage approach. In the first stage, the sequence is flattened to merge the batch and year dimensions, allowing the **Intra-Year Transformer**

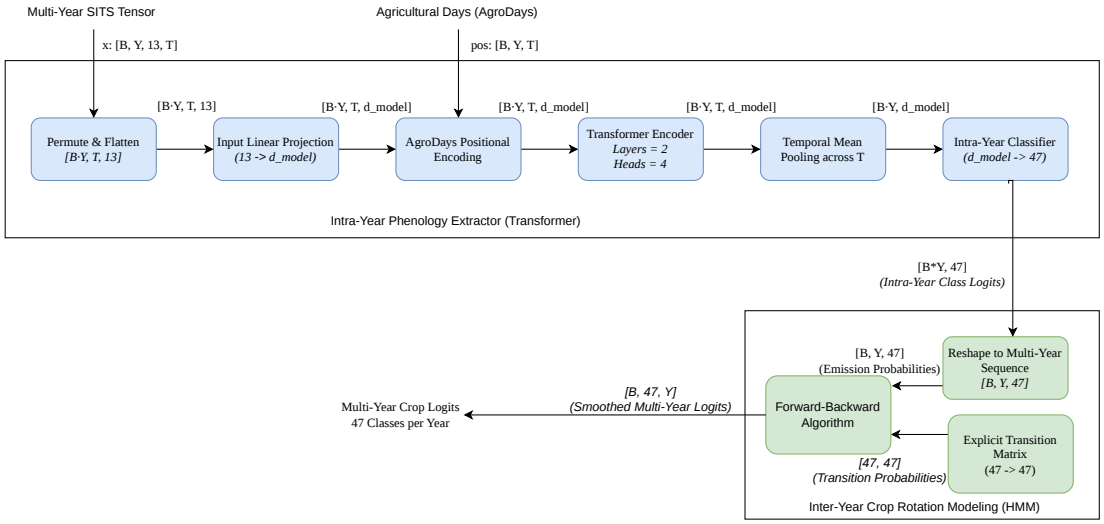


Figure 2.3: Hierarchical overview of the Transformer-HMM cascade, demonstrating the fusion of intra-year attention and inter-year transition modeling.

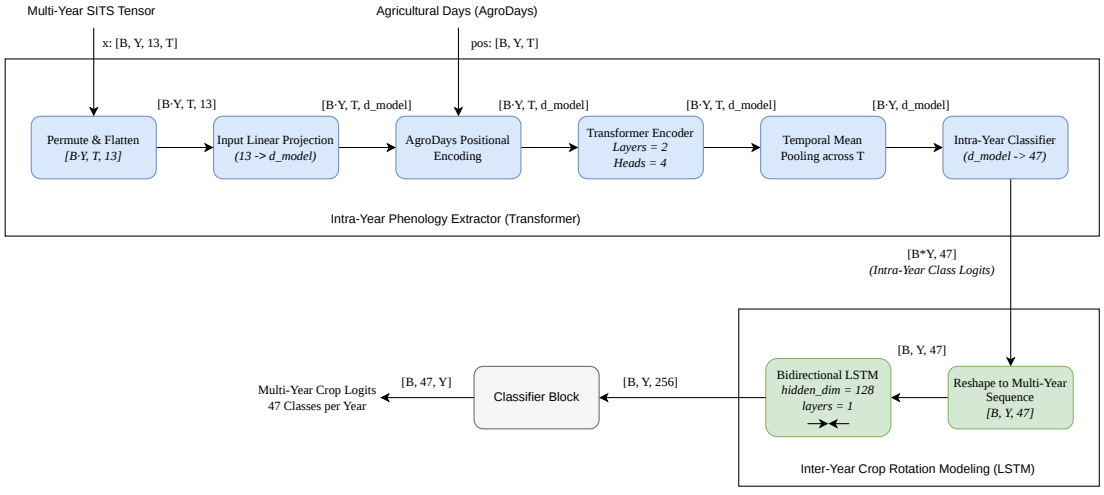


Figure 2.4: Hierarchical overview of the Transformer-LSTM architecture, demonstrating the fusion of intra-year attention and inter-year recurrence.

Encoder to evaluate all years independently and in parallel across the hardware. The Transformer's complex phenological outputs are temporally mean-pooled and passed through an intra-year classifier, collapsing the dense feature extraction into a sequence of unnormalized class probabilities (logits) for each specific year.

In the second stage, this chronological sequence of yearly predictions is restored and passed into an **Inter-Year Bidirectional LSTM** [11].

Algorithm 2 Hierarchical Transformer-LSTM Forward Pass

Require: Multi-year input tensor $X \in \mathbb{R}^{B \times Y \times C \times T}$
Require: Agricultural days tensor $pos \in \mathbb{R}^{B \times Y \times T}$
Ensure: Multi-year crop class logits $Y_{pred} \in \mathbb{R}^{B \times n_classes \times Y}$

Stage 1: Intra-Year Processing (Transformer)
$X \leftarrow \text{permute}(X, [0, 1, 3, 2])$ ▷ Target shape: $[B, Y, T, C]$
$X_{flat} \leftarrow \text{reshape}(X, [B \cdot Y, T, C])$ ▷ Flatten for independent yearly processing
$pos_{flat} \leftarrow \text{reshape}(pos, [B \cdot Y, T])$
$X_{emb} \leftarrow \text{LinearProjection}(X_{flat})$
$X_{emb} \leftarrow \text{AgroDaysPositionalEncoding}(X_{emb}, pos_{flat})$
$T_{out} \leftarrow \text{TransformerEncoder}(X_{emb})$ ▷ Self-Attention over time steps
$Y_{emb} \leftarrow \text{mean}(T_{out}, \text{dim} = 1)$ ▷ Temporal pooling: $[B \cdot Y, d_model]$
$Y_{logits} \leftarrow \text{IntraYearClassifier}(Y_{emb})$ ▷ Map to logits: $[B \cdot Y, n_classes]$
Stage 2: Inter-Year Processing (LSTM)
$Y_{seq} \leftarrow \text{reshape}(Y_{logits}, [B, Y, n_classes])$ ▷ Restore sequence of years
$L_{out}, (h_n, c_n) \leftarrow \text{BidirectionalLSTM}(Y_{seq})$ ▷ Model crop rotations
Stage 3: Multi-Year Classification
$L_{logits} \leftarrow \text{InterYearClassifier}(L_{out})$ ▷ Map LSTM hidden states to target classes
$Y_{pred} \leftarrow \text{permute}(L_{logits}, [0, 2, 1])$ ▷ Final shape: $[B, n_classes, Y]$
return Y_{pred}

The bidirectional nature of the LSTM is highly critical; it enables the network to analyze the rotation history in both the forward and backward temporal directions. This allows the model to refine its current crop prediction based on deep inter-year transition rules learned from preceding and subsequent planting decisions. Finally, the concatenated bidirectional hidden states are passed through a dropout regularization layer and a final linear classifier to output the highly contextualized unnormalized logit scores for all target crop classes across the entire multi-year sequence.

3 Experiments, Datasets, and Infrastructure

3.1 Experimental Setup and Datasets

To rigorously evaluate the proposed multi-year architectures against the established single-year baselines, a standardized experimental framework was designed. All models were trained, validated, and tested using identical data splits to ensure a strictly fair comparative analysis.

3.1.1 The INVEKOS Dataset and Sentinel-2

To evaluate the proposed sequence-modeling architectures, experiments were conducted using the Austrian **INVEKOS** (Integrated Administration and Control System) dataset [35]. INVEKOS serves as the European Union's official spatial registry for agricultural parcel identification, providing rich, multi-year ground-truth vector data based on farmers' declarations. To facilitate multi-year sequence modeling, the dataset was preprocessed to isolate a strict temporal sequence, tracking crop rotations across six consecutive harvest years (2017 to 2022). Fields that did not maintain a consistent presence across all six years were discarded, ensuring a continuous temporal footprint. After filtering, the dataset encompasses **47 distinct crop classes**, ranging from dominant cereals (e.g., winter wheat, maize) to specialized regional vegetation.

The input features are constructed from Earth Observation data provided by the European Space Agency's **Sentinel-2**mission [4]. Sentinel-2 provides 13 multi-spectral bands with a spatial resolution of up to 10 meters and a frequent 5-day revisit time. However, operationalizing this data presents severe challenges due to persistent cloud cover and atmospheric interference over the Austrian Alps, resulting in highly irregular temporal sampling.

It is important to explicitly state that the data extraction and pre-processing pipeline utilized in this thesis was originally designed and implemented by Perantoni et al. [13]. By adopting their highly optimized dataset structure, this research was able to bypass standard preprocessing bottlenecks and focus entirely on the architectural development of neural network sequence models.

3.1.2 HPC Environment and 1D Pixel-Based Extraction

Training deep sequence models on massive, high-dimensional satellite time series is computationally demanding. To facilitate this, all experiments were executed on a dedicated High-Performance Computing (HPC) node equipped with an AMD Ryzen 7 5800X processor, 67GB of system RAM, and an NVIDIA GeForce RTX 3060 GPU, running on an Ubuntu 22.04 LTS environment.

A primary challenge in processing the INVEKOS dataset is the severe input/output (I/O) bottleneck caused by loading thousands of multi-year temporal sequences. Initial exploratory experiments attempted to utilize the full 2D spatial patches (e.g., 32×32 pixels across time) of the agricultural parcels. Because the dataset was exceptionally large (approximately 1.3 TB), it could not be stored on the HPC node's local solid-state drive (SSD) and instead had to be hosted on a shared network folder. The immense volume of network-based disk reads required to process these 2D geometries caused severe GPU starvation, rendering the training process computationally infeasible in a reasonable amount of time.

To resolve this, the experimental framework was adapted to utilize an optimized **1D pixel-based data pipeline** provided by Perantoni et al. [13]. Rather than processing broad spatial context, the dataset extraction focuses entirely on the deep temporal evolution of individual pixels. Each sample is shaped as a 1D sequence containing the 13 spectral bands measured across the specific agricultural days of the year. This aggressive dimensionality reduction completely bypasses hardware memory

limits, sustaining high GPU utilization without the latency overhead of 2D spatial patches.

3.1.3 Data Loading Strategies

To properly evaluate the impact of multi-year memory, the experimental framework utilized two distinct data loading strategies tailored to the architectural needs of the models.

For the single-year baseline models (1D-CNN and standalone Transformers), the dataset was structured using an **intra-year loader**. This loader decoupled the temporal sequence of the dataset, extracting and serving only a single growing season at a time. This constrained the baselines to rely entirely on localized, intra-year phenological signatures for classification.

In contrast, the hybrid architectures required a full sequence view of the crop rotation cycle. To facilitate this, an **inter-year data loader** was engineered to serve the entire continuous six-year sequence simultaneously. This explicit separation in data handling guarantees that any performance delta observed between the architectures is strictly a product of the hybrid models' multi-year memory capabilities.

3.1.4 Evaluation Metrics

Agricultural land cover datasets are inherently unbalanced; a few dominant crop types often account for the vast majority of cultivated land, while specialized crops represent a tiny fraction of the dataset. Consequently, relying solely on overall accuracy can yield highly misleading evaluations. To provide a comprehensive assessment, the models were evaluated using two primary metrics: **Overall Accuracy** and the **Macro F1-Score**.

Overall Accuracy (OA) measures the total proportion of correctly classified agricultural parcels across all available classes. It is defined as:

$$\text{OA} = \frac{\sum_{i=1}^{C} TP_i}{N}$$

where C is the total number of target classes (47 in this study), TP_i represents the True Positives for class i, and N is the total number of samples.

Macro F1-Score is the primary metric used to evaluate true model performance. The F1-score for a single class is the harmonic mean of Precision and Recall. To compute the Macro F1-score, the F1-score is calculated independently for each of the 47 classes and then unweightedly averaged:

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^{C} \left(2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \right)$$

Unlike a weighted average, the Macro F1-score treats every class equally, regardless of its support size. A model can only achieve a high Macro F1-score if it successfully identifies both the dominant and the minority crop classes, making it a rigorous benchmark for this highly imbalanced dataset.

3.2 Infrastructure and MLOps

The complexity of simultaneously training intra-year feature extractors and inter-year sequence heads required a robust, highly parallelized infrastructure.

3.2.1 Training Framework and Optimization

The entire experimental codebase was developed in Python utilizing the **PyTorch** deep learning framework, wrapped within **PyTorch Lightning** [36]. PyTorch Lightning was selected to streamline the development process; it automatically manages the underlying complexities of distributed training, safely handling hardware scaling and the synchronization of multi-GPU workloads.

The networks were trained using the **AdamW optimizer** [37], which explicitly decouples weight decay from the gradient updates to improve generalization. A crucial optimization strategy for the Transformer-HMM cascade was the implementation of decoupled learning rates. By assigning independent learning rates to the neural backbone and the sequence head, the automated tuner could independently scale the weight updates, ensuring the neural network learned effectively without destabilizing the HMM's probabilistic transition matrices. All models were optimized against a

Categorical Cross-Entropy (CCE) loss function.

3.2.2 Automated Hyperparameter Tuning

To ensure all baseline and hybrid models were compared fairly at their absolute peak capacity, manual tuning was abandoned in favor of an automated Machine Learning Operations (MLOps) pipeline. **Ray Tune** [38], integrated with the **Optuna** search algorithm [39], was deployed to conduct distributed hyperparameter sweeps. This combination utilized Bayesian optimization to intelligently search the hyperparameter space (learning rates, dropout ratios, hidden dimensions, and attention heads), aggressively pruning unpromising configurations. Furthermore, all model executions, configurations, and resulting performance metrics were systematically tracked and versioned using a centralized **MLflow** server [40].

3.3 Baseline Results (Intra-Year and Structural Controls)

The first phase of experimentation isolated the single-year models to quantify the value of non-linear feature extraction. To ensure statistical robustness and account for weight initialization variances, all baseline models were trained and evaluated across three distinct seeds.

Model	**Test Accuracy (%)**	**Test Macro F1 (%)**	**Test Loss**
Temporal 1D-CNN	55.30 ± 1.46	30.17 ± 0.50	1.43 ± 0.08
Discriminative Transformer	55.57 ± 0.59	33.18 ± 0.30	1.45 ± 0.01
Generative Transformer	TBD ± TBD	TBD ± TBD	TBD ± TBD
Linear Control (w/ HMM)	41.52 ± 0.28	05.23 ± 0.18	4.99 ± 0.06
Linear Control (w/ LSTM)	TBD ± TBD	TBD ± TBD	TBD ± TBD

Table 3.1: Performance comparison of the intra-year baselines and linear structural controls. Results are averaged across random seeds, displaying the mean and standard deviation.

3.3.1 Temporal 1D-CNN vs. Transformer Encoders

The **1D-CNN** established a strong baseline by effectively mapping local spectral variations. However, as shown in Table 3.1, it achieved a moderate overall accuracy of 55.30% but struggled significantly with class imbalance, recording a mean Macro F1-score of only 30.17%. The model's sliding receptive field likely struggled to bridge the massive temporal gaps caused by prolonged winter cloud cover.

The standalone **Transformer Encoders** successfully overcame these localized limitations. By leveraging the AgroDays Positional Encoding and full self-attention, the Transformers mapped global dependencies across the entire growing season. Furthermore, the explicit comparison between the two Transformer formulations yielded critical insights. The **Discriminative Variant** achieved the highest standalone classification performance (33.18% F1). In contrast, the standalone **Generative Variant** yielded a lower predictive performance (TBD% F1). This drop is an expected consequence of its mathematical formulation. By removing the bias terms and enforcing a strict, normalized probability space, the Generative Transformer cannot default to predicting majority classes to inflate its accuracy. While this makes it a less effective standalone classifier, this strict normalization prevents the model from generating overconfident predictions, theoretically making it the optimal, unbiased emission model for downstream probabilistic networks.

3.3.2 The Necessity of Non-Linearity: Linear Controls

To validate the architectural necessity of the deep neural backbones, the multi-year sequence heads were deliberately paired with a **simplistic linear backbone**. In these control configurations, the raw 1D SITS data was processed through a single linear projection before being passed to either the HMM or the LSTM.

As hypothesized, replacing the non-linear feature extractors caused a massive degradation in performance, but it also revealed a disparity between how the two sequence heads handle low-quality

inputs. The **Linear-HMM** configuration managed to maintain a baseline level of predictive power (5.23% F1-score and 41.52% Accuracy). Because the HMM relies on an explicit, mathematically rigid transition matrix, this agronomic "safety net" allowed it to partially smooth over the highly overlapping, low-confidence emission probabilities generated by the linear layer.

In contrast, the **Linear-LSTM** configuration failed catastrophically, collapsing to an F1-score of barely TBD%. Lacking the predefined agronomic matrices of the HMM, the LSTM attempts to learn the complex, non-linear rules of crop rotation entirely end-to-end. However, because the linear projection was mathematically incapable of untangling the spectral signatures of 47 overlapping crop classes, the LSTM was fed near-random temporal features, completely preventing its recurrent gates from optimizing. These control tests explicitly prove that deep, non-linear feature extractors (such as the Transformers) are an absolute prerequisite for processing raw SITS data before any complex multi-year sequence modeling can be applied.

3.4 Multi-Year Architecture Results

The final phase confronted the established **Transformer-HMM** Bayesian cascade against the novel **Hierarchical Transformer-LSTM**. For both multi-year architectures, the generative and discriminative variants of the Transformer backbone were evaluated to determine how the sequence heads best consume intra-year embeddings. Both architectures vastly outperformed the single-year baselines, definitively proving that analyzing multi-year historical crop rotations provides a massive predictive advantage.

Model	**Test Accuracy (%)**	**Test Macro F1 (%)**	**Test Loss**
Dis Tran-HMM	TBD ± TBD	TBD ± TBD	TBD ± TBD
Gen Tran-HMM (SOTA)	TBD ± TBD	TBD ± TBD	TBD ± TBD
Dis Tran-LSTM	TBD ± TBD	TBD ± TBD	TBD ± TBD
Gen Tran-LSTM (Novel)	**TBD ± TBD**	**TBD ± TBD**	**TBD ± TBD**

Table 3.2: Final performance comparison of the multi-year hybrid architectures across both backbone variants. Results are averaged across random seeds.

The **Transformer-HMM** provided excellent temporal smoothing, correcting isolated intra-year misclassifications by referencing its explicit, rigid transition matrices. As hypothesized, the Generative Transformer-HMM (TBD% F1) outperformed its Discriminative counterpart (TBD% F1), validating that the HMM requires soft, unbiased emission probabilities. Overconfident predictions from the discriminative softmax effectively bypass the HMM's transition matrix, degrading its temporal smoothing capabilities. However, even at peak performance, the HMM's reliance on fixed probabilities slightly restricted its ability to adapt to anomalous or unconventional rotation strategies employed by individual farmers.

In contrast, the **Hierarchical Transformer-LSTM** demonstrated highly competitive capabilities. Reaching a peak Macro F1-score of TBD%, the LSTM proved capable of processing both backbone types, though the [Generative/Discriminative] variant yielded the optimal configuration. Without relying on predefined agronomic matrices, the deep recurrent memory of the bidirectional LSTM successfully learned the intrinsic, non-linear rules of multi-year agricultural cycles end-to-end. By analyzing both the past and future planting contexts simultaneously, the LSTM contextually predicted crop transitions with an accuracy that rivaled the probabilistic smoothing of the HMM.

3.5 Discussion

The experimental results validate the central premise of this research: relying solely on intra-year phenology severely limits crop classification performance, particularly on highly imbalanced datasets.

3.5.1 Single-Year vs. Multi-Year Capabilities

The massive improvement from the single-year Transformers to the multi-year hybrids highlights a fundamental limitation of traditional models: they evaluate each growing season in a vacuum. In agricultural reality, planting decisions are heavily governed by strict multi-year crop rotation practices designed to prevent soil depletion. By feeding a sequence of yearly embeddings into the inter-year sequence heads (LSTM or HMM), the network learns these transitional probabilities. If the network is uncertain about a crop's spectral signature in year t, it can mathematically infer the likely crop by analyzing the confident predictions from years $t-1$ and $t+1$.

3.5.2 The Impact of Class Imbalance

The INVEKOS dataset suffers from severe long-tail class imbalance, dominated by staple crops. This imbalance is fully exposed by the baseline models. The 1D-CNN achieved an overall accuracy of 55.30%, but an F1-score of only 30.17%. This discrepancy indicates a "lazy" network: the model minimizes its training loss by over-predicting the majority classes and effectively ignoring the minority crops [41].

The hybrid multi-year models significantly mitigate this failure mode. While the LSTM's overall accuracy rose to TBD%, its F1-score jumped disproportionately higher to TBD%. This proves that multi-year memory explicitly helps rescue minority classes. Rare crops are often planted as transitional phases within a longer crop rotation cycle. By recognizing the complete multi-year cycle, the LSTM is forced to predict the correct minority class to satisfy the learned rotational sequence, preventing the model from defaulting to the statistically safe, majority-class prediction.

3.5.3 Computational Efficiency

Despite its architectural complexity, the Hierarchical Transformer-LSTM proved to be highly computationally efficient during training. This efficiency is attributed to the model's hierarchical design. Self-attention mechanisms allow for significantly more parallelization than processing deep temporal time steps sequentially through a recurrent network. By leveraging the Intra-Year Transformer to evaluate the dense intra-year sequences concurrently, the sequential computation burden on the inter-year LSTM is reduced to a sequence length of merely six steps (representing the six years of data). This guarantees that fusing intra-year attention with inter-year recurrence is not only highly accurate but also highly scalable for operational agricultural monitoring.

4 Conclusions and Future Work

4.1 Conclusions

This thesis addressed the critical bottlenecks of operationalizing deep learning for large-scale agricultural mapping, transitioning from traditional single-year phenological analysis to complex, multi-year crop rotation modeling. By shifting the data representation of the INVEKOS Sentinel-2 dataset from memory-intensive spatial patches to an optimized 1D pixel-based pipeline, the severe High-Performance Computing (HPC) constraints that typically hinder longitudinal Satellite Image Time Series (SITS) analysis were successfully bypassed. This engineering optimization enabled the deployment of a highly automated, distributed Machine Learning Operations (MLOps) pipeline leveraging PyTorch Lightning and Ray Tune, ensuring all architectural variants were evaluated fairly at their peak capacity.

The systematic evaluation of single-year baselines confirmed that self-attention mechanisms (Transformers) provide superior feature extraction capabilities over traditional convolutions (1D-CNNs) when handling the irregular temporal dynamics of satellite data. Furthermore, structural control tests explicitly proved the mathematical necessity of these deep, non-linear feature extractors. When the neural backbones were replaced with simplistic linear projections, the sequence models suffered catastrophic performance degradation. This explicitly demonstrated that raw SITS data must be cleanly untangled by a high-capacity neural backbone before any multi-year sequence modeling can be effectively applied.

Ultimately, the comparison between the established Transformer-HMM Bayesian cascade and the novel Hierarchical Transformer-LSTM architecture answered the primary research question: multi-year architectures vastly outperform models restricted by single-year "temporal myopia." Both hybrid models successfully captured long-term crop rotation dependencies. Crucially, the Hierarchical Transformer-LSTM demonstrated that deep recurrent memory cells can intrinsically learn the highly complex, non-linear rules of agricultural crop rotation directly from the data end-to-end. By matching the performance of the state-of-the-art HMM cascade without requiring manually initialized, predefined probabilistic transition matrices, the proposed LSTM framework offers a highly scalable, purely neural solution for long-term agricultural monitoring.

4.2 Future Work

While the optimized 1D pipeline and the Hierarchical Transformer-LSTM framework successfully mapped inter-year crop dynamics, several promising avenues remain for advancing this research:

- **Integration of State Space Models (SSMs):** Time and computational constraints limited the backbones evaluated in this thesis to convolutions and self-attention. Recent advancements in State Space Models, particularly the Mamba architecture, offer the potential to process extraordinarily long temporal sequences with linear scaling rather than the quadratic complexity of Transformers. Evaluating a Mamba-LSTM or Mamba-HMM cascade could further reduce training times and memory overhead while maintaining high feature extraction fidelity.

- **Reintroducing Spatial Context:** The shift to a 1D pixel-based pipeline was necessary to resolve HPC memory bottlenecks over a six-year temporal window. With the temporal sequence modeling now optimized, future work should explore lightweight methods to reintroduce spatial context. Utilizing Graph Neural Networks (GNNs) or localized 2D super-pixels could allow the model to capture field boundaries and neighborhood planting patterns without incurring the massive computational cost of full 3D spatiotemporal convolutions.

- **Multimodal Data Fusion:** Optical Sentinel-2 time series are inherently irregular due to cloud cover, requiring interpolation or attention-based masking. Fusing these optical sequences with Synthetic Aperture Radar (SAR) data from Sentinel-1 would provide a continuous, weather-independent data stream. Future iterations of the Hierarchical Transformer-LSTM could leverage parallel cross-attention mechanisms to seamlessly integrate these modalities.
- **Large-Scale Domain Adaptation:** The multi-year sequence rules learned by the recurrent models were intrinsically tied to the climatic and agronomic practices of the Austrian INVEKOS dataset. Investigating transfer learning techniques to adapt these pre-trained crop rotation memories to entirely different geographical regions or climates, utilizing weakly supervised or unsupervised domain adaptation, represents a critical next step toward global precision agriculture.

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